Day 045 | Unit 03 | Teacher plan

Integers and Rational Numbers

Unit 3 Test

Standards: 6.NS.A.1, 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.7c, 6.NS.C.7d, 6.NS.C.6b, 6.NS.C.8

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

Assessment routine

5 minutes: explain response types without solving any item. 35 minutes: independent assessment. 10 minutes: collect work and note misconceptions for the next lesson. Keep the answer key and worked-example help unavailable during assessment.

Review allocation

Questions 1–4 are previous-unit review (20%). Current-unit questions increase in demand after the review. Unit 1 uses Grade 5 readiness skills because no earlier course unit exists.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 1, 7, 15, 22, 24

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Unit 3 Test

Standards: 6.NS.A.1, 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.7c, 6.NS.C.7d, 6.NS.C.6b, 6.NS.C.8

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

Directions

The first 4 questions (20%) review Unit 2. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Calculate 3/4 × 1/2. Simplify.

02 / Previous unit review | Constructed response

Calculate 3 1/4 × 1 1/3. Simplify.

03 / Previous unit review | Constructed response

Draw a bar partitioned into 12 equal parts. Shade 11 parts. Circle groups of 7 parts. Use the model to find 11/12 ÷ 7/12.

04 / Previous unit review | Constructed response

Draw a bar partitioned into 6 equal parts. Shade 6 parts. Circle groups of 3 parts. Use the model to find 1 ÷ 1/2.

05 / Constructed response

A marker is 8 m above sea level. Write its signed elevation and explain what elevation 0 means.

06 / Constructed response

Find the opposite of -8. Show both values on a number line.

07 / Constructed response

Draw a horizontal or vertical number line. Plot and label -3, 0, and 5.

08 / Constructed response

Draw a horizontal or vertical number line. Plot and label -8/3, 0, and 4/3.

09 / Constructed response

Draw a horizontal or vertical number line. Plot and label -1, 0, and 3/2.

10 / Constructed response

Compare -4/5 ___ -1 using <, >, or =. Sketch a number line to justify.

11 / Constructed response

Compare -4/5 ___ -7/10 using <, >, or =. Sketch a number line to justify.

12 / Multiple select

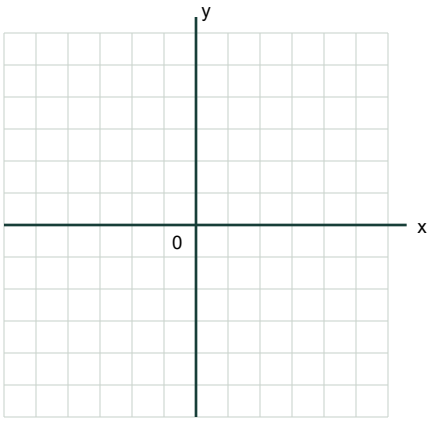
Consider -8 and 6. Select all true statements.
A. -8 > 6
B. |-8| = -8
C. |-8| = 8
D. -8 is farther from zero than 6.

13 / Multiple select

Consider -5 and 3. Select all true statements.
A. |-5| = -5
B. |-5| = 5
C. -5 is farther from zero than 3.
D. -5 > 3

14 / Coordinate plotting and explanation

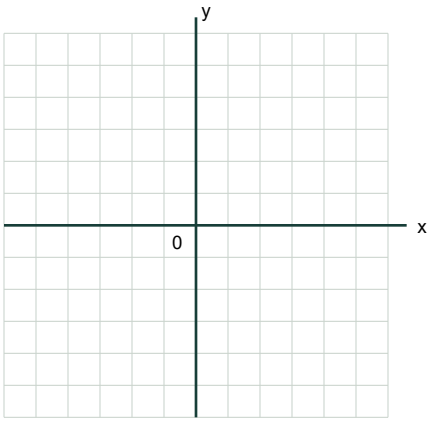
Plot P(-2, 6), Q(2, -6), and R(0, -6). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

15 / Coordinate plotting and explanation

Plot P(-3, 5), Q(3, -5), and R(0, -5). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



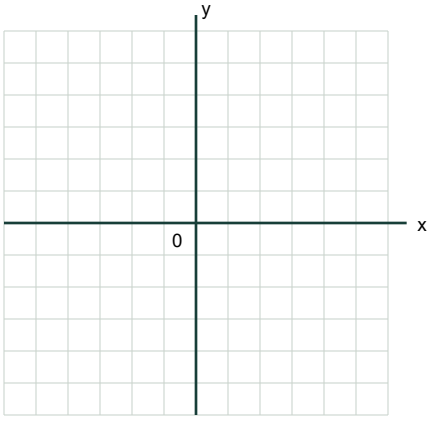
Label the scale and units.

16 / Multiple select

Point P is (-6, 3). Select all correct reflections.
A. Across the y-axis: (6, 3).
B. Across the x-axis: (6, 3).
C. Across both axes: (-6, 3).
D. Across the x-axis: (-6, -3).

17 / Coordinate plotting and numeric entry

Plot A(-5, -6), B(2, -6), and C(-5, 6). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

18 / Multiple select

Point P is (-5, 5). Select all correct reflections.
A. Across the x-axis: (-5, -5).
B. Across the x-axis: (5, 5).
C. Across both axes: (-5, 5).
D. Across the y-axis: (5, 5).

19 / Multiple select

A sensor is 7 meters below a reference point. Select all correct interpretations.
A. Its signed position is -7 m.
B. Its signed position is +7 m.
C. The opposite position is 7 m below the reference.
D. A reading of 0 m means at the reference point.

20 / Multiple select

Select all true statements about -7/2 and its opposite. Explain the role of zero.
A. The opposite of the opposite of -7/2 is -7/2.
B. The opposite of 0 is 1.
C. -7/2 and 7/2 are on the same side of zero.
D. The opposite of -7/2 is 7/2.

Day 045 | Unit 03 | Answer key

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Unit 3 Test

Standards: 6.NS.A.1, 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.7c, 6.NS.C.7d, 6.NS.C.6b, 6.NS.C.8

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 3/8

3/4 × 1/2 = 3/8.

6.NS.A.1 | Constructed response

2. 13/3

13/4 × 4/3 = 13/3.

6.NS.A.1 | Constructed response

3. 11/7

Each small part is 1/12; 11 shaded parts divided into groups of 7 gives 11/7 = 11/7 groups.

6.NS.A.1 | Constructed response

4. 2

Each small part is 1/6; 6 shaded parts divided into groups of 3 gives 6/3 = 2 groups.

6.NS.A.1 | Constructed response

5. +8 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

6. 8

-8 and 8 are each 8 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

7. -3 < 0 < 5; unit steps

Negative -3 is left of or below 0; 5 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

8. -8/3 < 0 < 4/3; use steps of 1/3

Negative -8/3 is left of or below 0; 4/3 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

9. -1 < 0 < 3/2; use steps of 1/4

Negative -1 is left of or below 0; 3/2 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

10. >

-4/5 > -1; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

11. <

-4/5 < -7/10; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

12. C, D - |-8| = 8; -8 is farther from zero than 6.

Absolute value measures distance: 8 versus 6. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

13. B, C - |-5| = 5; -5 is farther from zero than 3.

Absolute value measures distance: 5 versus 3. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

14. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

15. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

16. A, D - Across the y-axis: (6, 3).; Across the x-axis: (-6, -3).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (6, -3).

6.NS.C.6b | Multiple select

17. AB = 7 units; AC = 12 units

AB is horizontal: 5 + 2 = 7. AC is vertical: 6 + 6 = 12. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

18. A, D - Across the x-axis: (-5, -5).; Across the y-axis: (5, 5).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (5, -5).

6.NS.C.6b | Multiple select

19. A, D - Its signed position is -7 m.; A reading of 0 m means at the reference point.

Below the chosen zero is negative, so the position is -7. Zero identifies the reference; the opposite is 7 m above it.

6.NS.C.5 | Multiple select

20. A, D - The opposite of the opposite of -7/2 is -7/2.; The opposite of -7/2 is 7/2.

An opposite reverses direction without changing distance. Reversing twice returns to -7/2. Zero is its own opposite.

6.NS.C.6a | Multiple select